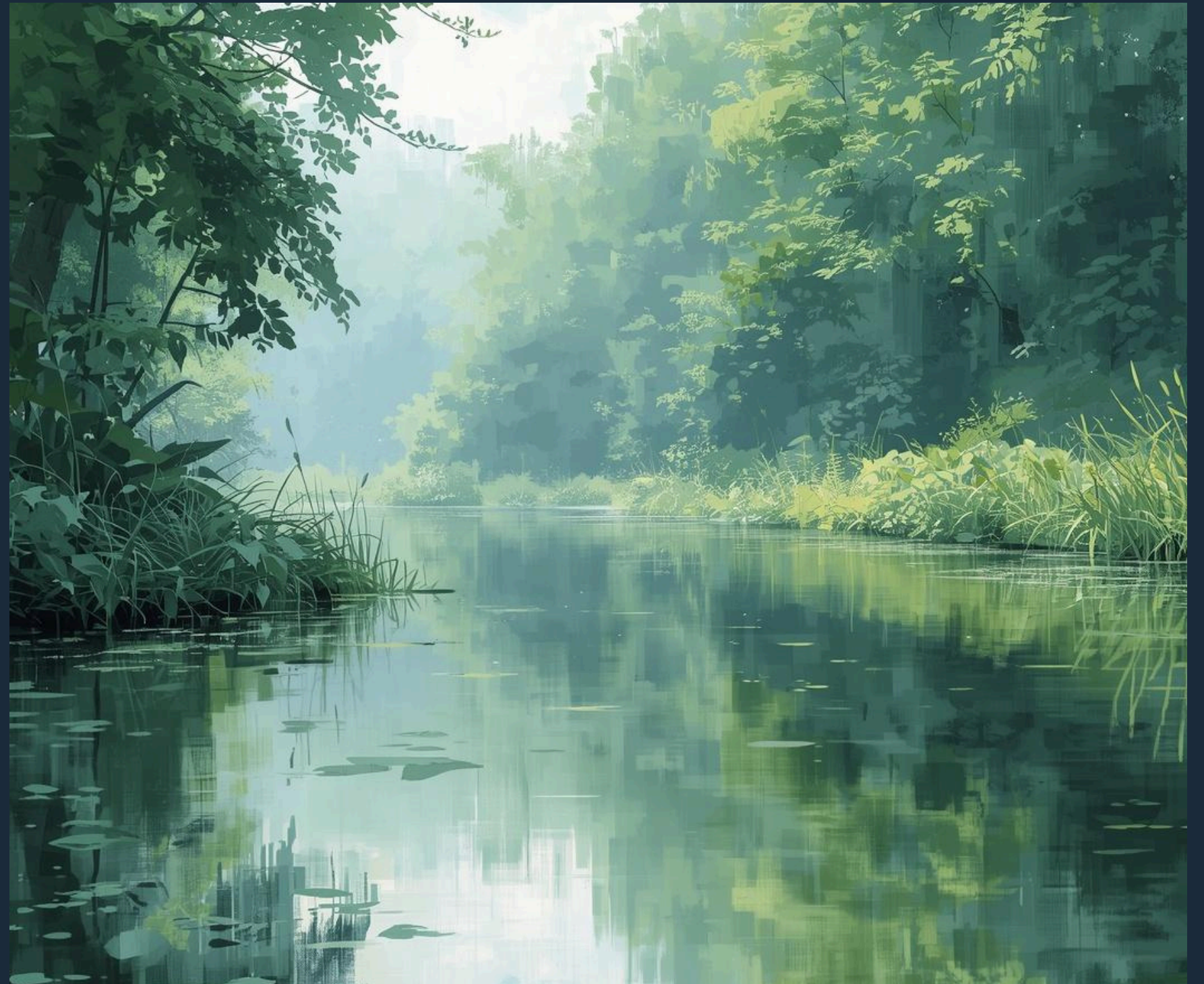


Weather Patterns and Malaria Incidence in Nigeria

COURSE: DATA ANALYSIS AND
VISUALIZATION (3MTT NEXTGEN)

STATES: FCT, NASARAWA,
NIGER, KOGI, KADUNA

Dorathy Chinenye Ugwuoke



Project Introduction

Analyzing the relationship between weather and malaria incidence in Nigeria

The project aims to explore the **interconnection** between various weather patterns, such as rainfall, temperature, and humidity, and their impact on malaria incidence across selected Nigerian states.

By identifying **patterns and correlations**, the findings will provide crucial insights to enhance public health strategies and improve malaria prevention efforts.



Problem Statement: Malaria and Environmental Conditions



Malaria Outbreaks

Environmental conditions significantly influence malaria transmission patterns.



Dataset Overview of Malaria Analysis

This dataset provides crucial insights into malaria cases and environmental factors across Nigeria, including rainfall, temperature, and humidity, to better understand the relationship between weather patterns and malaria incidence.

Dorathy's Malaria_Weather_Correlation Dataset ☆ 📁 ☁						
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	A	B	C	D	E	F
1	Location	Month	Malaria_Cases	Avg_Temperature(C)	Rainfall(mm)	Humidity(%)
2	FCT	2023-01	429	31.5	247	61.5
3	FCT	2023-02	594	31.6	307.8	81.2
4	FCT	2023-03	209	28	333.3	65.2
5	FCT	2023-04	437	24.2	235.3	87.6
6	FCT	2023-05	242	30.8	241.5	77.4
7	FCT	2023-06	271	30.3	102.2	72.3
8	FCT	2023-07	593	26.4	101.7	87.1
9	FCT	2023-08	281	29.1	149.9	65.7
10	FCT	2023-09	422	31.5	69.7	72.1
11	FCT	2023-10	311	28.3	287.3	64.2
12	FCT	2023-11	260	28.3	54.4	85
13	FCT	2023-12	289	28.3	183.3	68.5
14	Nasarawa	2023-01	198	28.8	303	83.9
15	Nasarawa	2023-02	194	27.8	92.6	77.8
16	Nasarawa	2023-03	253	24.2	42.2	63
17	Nasarawa	2023-04	275	31.5	191.7	72.1

Tools and Methodology Overview



Analytical Approach

Our process includes data cleaning, analysis, and visualization.



Tools & Methodology

Analyzing Malaria and Weather Data

Data Collection

Data was sourced from the **National Bureau of Statistics** and **NIMET**, ensuring comprehensive coverage of malaria cases across relevant states for accurate analysis.

Data Cleaning

The dataset underwent thorough cleaning to remove inconsistencies, missing values, and outliers, ensuring the integrity and reliability of the data for further analysis.

Correlation Analysis

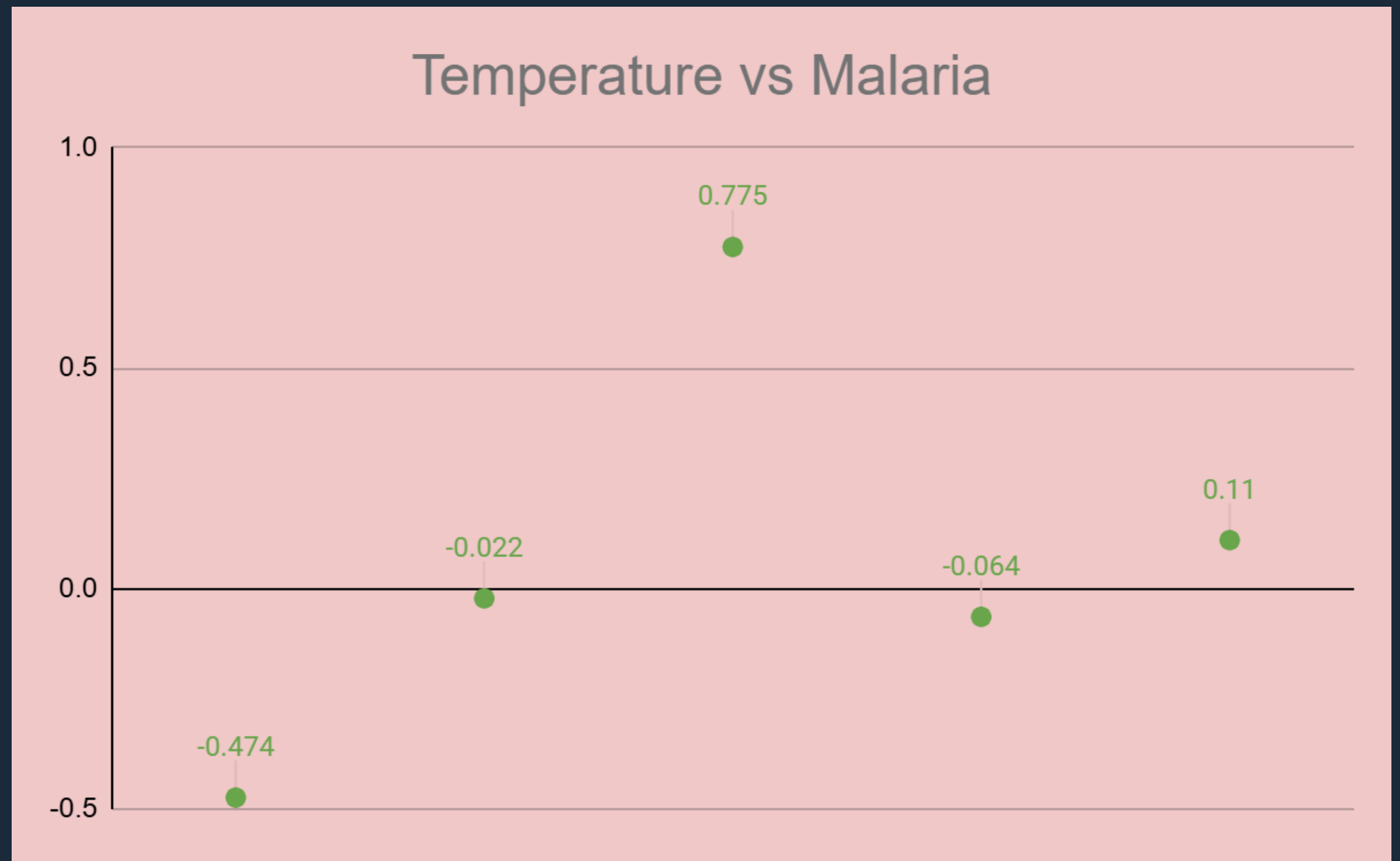
Correlation analysis was performed using statistical methods to identify relationships between weather variables and malaria incidence, providing insights into potential predictive factors for outbreaks.



Temperature and Malaria Correlation

Positive Correlation

Increased temperature relates to higher malaria cases observed.

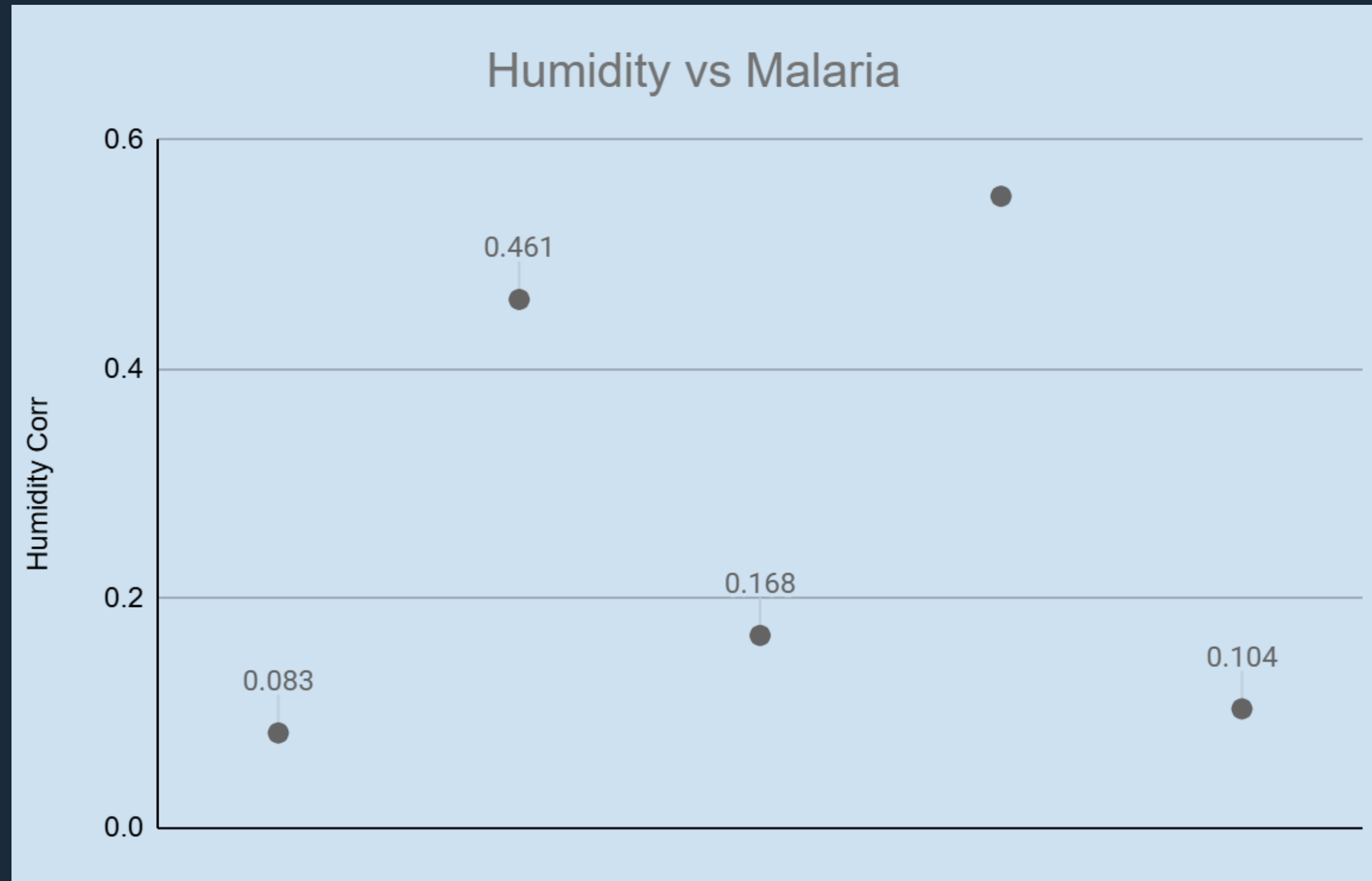


Here we see a strong positive relationship between Temperature and malaria cases in Niger State.

Humidity's Impact on Malaria Incidence in Kaduna

Humidity Influence

Increased humidity levels contribute to higher malaria cases.



Humidity shows a strong influence in Kaduna, suggesting that moist conditions support malaria transmission

Summary of Insights

Malaria Correlation Summary – Insights				
Summary Table Layout ▾				
State ▾	Temp Corr ▾	Rainfall Corr ▾	Humidity Corr ▾	Key Insight ▾
Kogi	-0.474	-0.265	0.083	Moderate negative correlation with temperature; humidity minimal influence.
FCT	-0.022	-0.004	0.461	Malaria cases moderately influenced by humidity; temperature and rainfall weak.
Niger	0.775	-0.22	0.168	Very strong positive correlation with temperature; malaria increases in hotter periods.
Kaduna	-0.064	-0.4	0.551	Strong positive correlation with humidity; malaria risk rises during humid periods.
Nasarawa	0.11	-0.055	0.104	Weak correlations overall; no strong weather predictor identified.

This analysis highlights the **key environmental factors** impacting malaria cases across various states. Consistently, **humidity emerges as the most significant** predictor of malaria incidence, while temperature notably affects cases in Niger State. These insights provide a foundation for targeted public health interventions and effective resource allocation.



GitHub Demo

Interactive Interface for Malaria Prediction Using Weather Data



Malaria Risk Dashboard

AI-powered outbreak prediction using weather data

Weather Input

Rainfall (mm)

Temperature (°C)

Humidity (%)

 Predict Risk

Prediction Result

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Trend Forecast

The Streamlit application serves as a user-friendly interface that allows stakeholders to **explore relationships** between weather variables and malaria incidence.

By inputting various weather data, users can visualize potential malaria outbreaks, enhancing predictive capabilities and supporting public health planning in Nigeria. This tool is a step towards data-driven decision-making.

Impact

Enhancing Public Health through Data-Driven Insights

The analysis of weather patterns and malaria incidence offers **significant benefits**. By identifying correlations between environmental factors and malaria outbreaks,

we can improve prediction capabilities, allocate health resources effectively, and enhance public health planning.

This proactive approach facilitates targeted interventions and supports community health resilience against malaria.



Recommendations

Proposals for Predictive Models and Partnerships in Malaria Prevention

Building a **predictive model leveraging weather data** can enhance malaria incidence forecasting.

This approach should be complemented by actionable solutions focused on prevention and control.

Forming partnerships with health agencies will ensure effective implementation, ultimately leading to improved public health outcomes and better resource allocation in response to malaria outbreaks.



Conclusion

Understanding weather's impact on malaria incidence in Nigeria

In summary, our analysis reveals that **weather variables**, particularly humidity and temperature, significantly influence malaria incidence across Nigeria.

By leveraging **data-driven insights**, we can enhance public health strategies, improve intervention targeting, and ultimately, reduce malaria outbreaks.

Thank you for your attention and interest in this critical issue.

